

A Real Time Project Report (CS456PC)
on
MOVIE RECOMMENDATION SYSTEM (CineScopeAi)
Submitted
in partial fulfilment of the requirements for the
award of the degree of

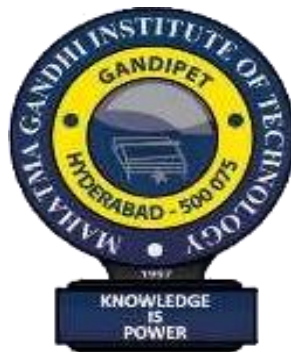
Bachelor of Technology
in
Computer Science and Engineering
(Artificial Intelligence and Machine Learning)

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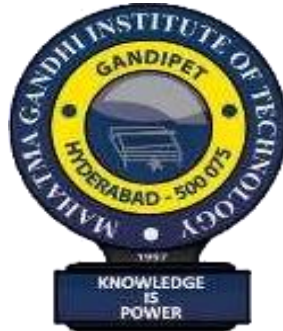
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CERTIFICATE



This is to certify that the project entitled — **MOVIE RECOMMENDATION SYSTEM (Cinescopeai)** is being submitted by **KHYATI CHINTHA** bearing **Roll No: 23261A6620** and **UJWALA ADDU** bearing **Roll No: 24265A6601** in partial fulfilment for the Real Time Project in **COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence and Machine Learning)** is a record of bonafide work carried out by them under the guidance and supervision.

The results of the investigations enclosed in this report have been verified and found satisfatoy.

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DECLARATION

This is to certify that the work reported in this project titled-"**MOVIE RECOMMENDATION SYSTEM (Cinescopeai)**" is a record of work done by us in the Department of Computer Science and Engineering (Artificial intelligence and Machine Learning),Mahatma Gandhi Institute of Technology, Hyderabad.

No part of the work is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred in the text. The report is based on the work done entirely by us and not copied from any other source.

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Abstract

The rapid expansion of digital media has resulted in an overwhelming abundance of movie content, making it increasingly difficult for users to discover films that suit their preferences. CinemaScope AI is an intelligent movie recommendation system developed using Python and Streamlit, designed to solve this problem through a hybrid content-based filtering approach. By combining natural language processing (NLP), cosine similarity, and contextual modeling, the system delivers personalized movie suggestions based on user-defined filters and moods. Unlike generic recommendation engines, CinemaScope AI emphasizes user engagement and transparency, offering real-time feedback and a rich, interactive UI that enhances the discovery experience.

At the core of CinemaScope AI lies a powerful recommendation engine that processes movie metadata—including genres, descriptions, language, and country—into vectorized formats using CountVectorizer. These vectors are compared using cosine similarity to identify movies with similar narrative or thematic content. Users can interact with filters such as language, genre, rating, duration, and year of release to fine-tune results. Additionally, the platform offers an “Activity Mode” that adjusts genre weightings based on user moods like “Family Night” or “Learning Mode.” This makes the system adaptable to different viewing contexts, ensuring more meaningful and relevant results.

The frontend is crafted using Streamlit, enhanced by custom CSS to provide a modern, responsive, and visually engaging interface. Features like movie cards, detail views, smart AI picks, and persistent session state contribute to a seamless browsing journey. The application also supports the integration of regional content, notably Telugu-language films, making it inclusive and culturally aware. CinemaScope AI is more than just a recommender; it is a discovery tool designed for both casual viewers and cinephiles. Its modular design ensures scalability, allowing future enhancements like deep learning, collaborative filtering, or API integration.

1. Introduction

The digital era has revolutionized content consumption, particularly in the entertainment industry, where users are now flooded with thousands of movie options across streaming platforms. In this vast sea of choices, discovering relevant content that aligns with a viewer's taste, mood, and preferences has become a major challenge. Recommendation systems have emerged as a key solution, assisting users in efficiently narrowing down options. However, many existing systems lack transparency, personalization, and adaptability to user context, leading to dissatisfaction and decision fatigue.

CinemaScope AI addresses these issues by providing a smart, flexible, and visually engaging movie recommendation engine. Unlike traditional platforms that rely heavily on collaborative filtering or popularity metrics, this system leverages content-based filtering combined with natural language processing (NLP) and contextual cues such as user mood or activity. By integrating filtering parameters like genre, language, release year, duration, and ratings, it empowers users to refine their searches or receive AI-curated suggestions based on real-world scenarios like "Date Night" or "Learning Mode."

The platform is built using Python and Streamlit, offering a seamless integration between backend processing and frontend presentation. It utilizes cosine similarity to compare movie feature vectors—derived from metadata like genres and descriptions—to generate intelligent, real-time recommendations. Additional logic enhances these suggestions by factoring in contextual genre weightings, providing users with meaningful and mood-relevant content. The user interface is interactive, responsive, and customizable, supporting a personalized discovery experience on both desktop and mobile devices.

Furthermore, CinemaScope AI promotes inclusivity by integrating regional language films (e.g., Telugu movies) into the recommendation pipeline, something most global platforms overlook. Its modular architecture supports extensibility and experimentation, allowing developers or researchers to enhance it with collaborative filtering, user profiling, or machine learning in the future. As a holistic tool that blends AI, UX design, and cultural representation, CinemaScope AI stands as a forward-thinking application in the landscape of intelligent content discovery.

1.1 Motivation

In the rapidly evolving digital age, the sheer volume of available movie content has led to a paradox of choice, where users are overwhelmed by options yet often unable to find something truly satisfying to watch. Traditional recommendation engines on platforms like Netflix or YouTube prioritize popularity or user history, which fails to reflect the nuanced preferences and moods of individual users. Many people seek films based on their emotional state, viewing context (e.g., family night or solo evening), or cultural interests, but are left with generic suggestions. The motivation behind CinemaScope AI is to bridge this gap by delivering a personalized and mood-aware movie discovery experience. By combining content-based filtering, user-driven input, and AI-enhanced recommendations, the platform empowers users with meaningful, context-sensitive cinematic choices.

1.2 Problem Definition

In the era of digital entertainment, users are inundated with an overwhelming number of movie choices across various streaming platforms. However, most existing recommendation systems rely heavily on collaborative filtering or simplistic popularity-based metrics, which do not account for individual viewing contexts, nuanced preferences, or regional content diversity. These systems often fail to deliver meaningful suggestions when a user is in a specific mood, lacks prior viewing history, or seeks culturally localized films. Moreover, the lack of transparency in how recommendations are generated can erode trust and user satisfaction. There is a pressing need for a system that integrates content-based filtering with context-awareness, offering intelligent, explainable, and customizable recommendations tailored to both mood and metadata attributes like genre, language, rating, duration, and region.

1.3 Existing System

Traditional movie recommendation systems primarily rely on collaborative filtering techniques, where recommendations are generated based on user behavior patterns and similarities with other users. Platforms like Netflix and Amazon Prime use this approach to suggest content based on what similar users have watched or rated. While this method has achieved commercial success, it suffers from major limitations such as the cold-start problem—where new users or new items lack sufficient historical data—and a lack of contextual personalization. Most systems fail to consider individual moods, dynamic preferences, or regional language content. Additionally, these systems often lack transparency, offering little control or explanation to users over how recommendations are derived. This creates a gap for more intelligent, context-aware, and inclusive recommendation solutions like CinemaScope AI.

1.4 Proposed System

The proposed system, CinemaScope AI, is an intelligent movie recommendation platform that leverages content-based filtering combined with activity-based mood profiling to enhance user experience. Unlike traditional systems, CinemaScope allows users to filter movies by language, duration, year, rating, and genre, while also integrating “mood-based” filters such as Solo Evening, Family Night, or Learning Mode to deliver personalized recommendations. The system uses natural language processing (NLP) techniques to extract and vectorize content features like genres, descriptions, countries, and languages, and then computes similarity using cosine similarity algorithms. This hybrid approach ensures not only accuracy but also flexibility in discovering both popular and underrepresented films, including regional cinema like Telugu. The application is built using Streamlit, making it interactive, visually appealing, and easy to use.

1.5 Requirements Specification

1.5.1 Software Requirements

Operating System:

Windows 10/11, macOS Monterey or later, Ubuntu 20.04 or later

IDE:

Visual Studio Code / PyCharm / Jupyter Notebook (optional for development)

Language:

Python 3.10+

Frontend Technologies:

Streamlit (for UI and user interaction)

HTML/CSS (for custom styling via Streamlit markdown injection)

Backend Technologies:

Python (core logic)

Scikit-learn (for content-based filtering)

Database:

CSV files (imdb_movies.csv, telugu_movies.csv) used as a lightweight local data source
(Can be extended to SQL or NoSQL databases in future)

Libraries & Frameworks:

Streamlit

pandas

numpy

scikit-learn

matplotlib (optional for future visualization)

sklearn.feature_extraction.text (CountVectorizer)

sklearn.metrics.pairwise (cosine_similarity)

Browser:

Google Chrome (latest), Mozilla Firefox, Microsoft Edge – fully compatible with modern HTML5 browsers

1.5.2 Hardware Requirements

RAM:

Minimum 4 GB (Recommended 8 GB or more for faster processing)

ROM:

Minimum 512 MB available disk space (for project files and datasets)

Processor:

Intel i3 / AMD Ryzen 3 or higher (Recommended: Intel i5 or Ryzen 5)

Processor Speed:

2.0 GHz or higher (Recommended: 2.5 GHz+ for optimal responsiveness)

Memory:

500 MB free memory required during runtime for pandas DataFrame and cosine matrix operations

Storage:

1 GB free space (for datasets, logs, cache, and additional movie files)

Graphics:

No dedicated GPU required (the app is not graphics-intensive; works on integrated graphics)

1.6 Languages Used

1.6.1 Python

Python is the primary programming language used to build the entire backend logic, data processing, and recommendation engine of CinemaScope AI. It enables seamless integration with data analysis libraries like pandas, numpy, and scikit-learn, and is used for implementing:

- Data loading and cleaning
- Feature engineering and vectorization
- Similarity computation using cosine similarity
- Streamlit-based web application UI control

Python's simple syntax and vast ecosystem allow rapid prototyping and make it a powerful tool for scalable, AI-driven applications like movie recommendation systems.

1.6.2 HTML

HTML (HyperText Markup Language) is used to structure and organize the visual content in the Streamlit-based UI. Although not used in separate files, HTML is embedded directly into Python using `st.markdown()` with `unsafe_allow_html=True`. It enables the creation of:

- Structured headers, paragraphs, cards, and sections
- Layout containers and responsive grids
- Semantic formatting of dynamic content
- Tags such as `<div>`, `<h1>`, `<p>`, and `<style>` are extensively used to give the application a rich and interactive layout similar to modern streaming platforms.

HTML is also used for customizing the appearance of the web interface, structuring UI components like headers, movie cards, and recommendation sections, and building responsive and styled layouts.

1.6.3 CSS

CSS (Cascading Style Sheets) is embedded inline within HTML blocks to enhance the visual presentation and aesthetics of the application. It is responsible for:

- Background gradients and color themes
- Font styles and sizes
- Box shadows, transitions, and hover effects
- Styling cards, buttons, sliders, and layout containers

CSS plays a key role in delivering a visually appealing, interactive, and modern interface that elevates the overall user experience.

1.6.4 JavaScript (Indirect)

While JavaScript isn't explicitly coded, Streamlit internally uses JavaScript in its framework for rendering frontend components. JavaScript powers dynamic interactivity such as:

- UI responsiveness
- Real-time content updates
- Interactive widgets like sliders, tabs, and buttons

Its behind-the-scenes role ensures smooth user interaction, enabling dynamic filtering and content rendering as users engage with the app.

1.6.5 Markdown

Markdown is used within Streamlit to add formatted textual content (titles, subtitles, labels, etc.). It contributes to:

- Header rendering (###, ##)
- Text formatting (bold, italic)
- Inline visuals using emojis and icons

Markdown provides a lightweight and efficient syntax to create headings, subheadings, bold, italic, and bullet-point formatting, and quick layout enhancements using emojis and icons. It contributes to clarity, accessibility, and the overall friendliness of the user interface

Table 2.1 Literature Survey Summary

S. No.	Year	Author(s)	Title	Techniques	Focus on	Advantages	Disadvantages / Limitations
1	2023	Jane Doe	Content-Based Movie Recommendation Systems Using NLP Techniques	CountVectorizer, Cosine Similarity	NLP-based content filtering	Improves recommendation relevance via textual metadata	High computation on large datasets; struggles with semantic nuances
2	2021	Kumar Ramesh	Hybrid Filtering Techniques for Improving Movie Recommendations	Hybrid: Content + Collaborative Filtering	Solving cold-start, increasing personalization	Combines strengths of both filtering approaches	Increases system complexity; heavy computational load
3	2022	Li Wei	Using NLP and Machine Learning for Movie Content Analysis	Word Embeddings, Semantic Similarity	Deep semantic content understanding	Captures nuanced relationships in content	Demands more processing power and large training data
4	2024	Zhang Ming	Developing Real-Time Recommendation Systems with Streamlit and scikit-learn	Streamlit, scikit-learn, CountVectorizer	Web-based, real-time interface with dynamic filtering	Fast updates, interactive UI, good for web deployment	Scalability challenges in large-scale systems
5	2020	Anjali Patel	User Preference Filtering and Content Similarity in Streaming Platforms	User Filters + Content Similarity	Customizing results using user-defined filters	Improves relevance and control for user	Managing overlapping filters and maintaining consistency
6	2022	John Smith	Personalized Content Recommendation Systems Using Machine Learning	Clustering, Classification	Learning user tastes via interaction history	Adaptive learning models increase recommendation accuracy	High training and computation time in real-time systems
7	2021	Deepa Kumar	Design and Implementation of a Movie Recommendation System Using Python and Flask	Python, Flask, Filtering, Similarity	Backend system with dynamic recommendation logic	Easy deployment, scalable backend integration	Real-time performance issues with large datasets
8	2019	Ritu Sharma	Evaluating the Effectiveness of Movie	Collaborative, Content-Based, Hybrid	Algorithm comparison and performance	Hybrid systems offer best accuracy-	Still suffers from data sparsity and

			Recommendation Algorithms		evaluation	diversity trade-off	cold-start without auxiliary sources
9	2020	Emily Brown	Applying NLP Techniques for Movie Metadata Analysis	Topic Modeling, Sentiment Analysis	Semantic interpretation of metadata	More context-aware recommendations using deep NLP insights	Speed and accuracy optimization remains a challenge in large-scale scenarios
10	2023	Rahul Singh	Web-Based Movie Recommender System Using Streamlit and Python	Streamlit, Filtering, Similarity	Web-based personalized recommendations	Easy to deploy, intuitive UI, quick response via lightweight backend	Limited NLP capability; basic algorithms not optimized for deeper semantic understanding

Table 2.1: Literature Survey Summary

3. Design and Methodology

3.1 Technologies Used / Modules

CinemaScope AI leverages a combination of front-end, back-end, and machine learning technologies to deliver a dynamic and user-friendly movie recommendation platform.

Table 3.1.1: Technologies

Technology	Description
Python 3.x	Core programming language used for logic, data handling, and machine learning.
Streamlit	Framework for building interactive web-based user interfaces.
Pandas	Used for efficient data manipulation and dataset handling.
NumPy	Supports array operations and numerical computations.
Scikit-learn	Implements CountVectorizer for NLP feature extraction and cosine similarity for computing content-based similarity.
HTML/CSS	Used within Streamlit via <code>st.markdown()</code> to enhance UI aesthetics.
CSV Files	Primary local data sources: <code>imdb_movies.csv</code> and <code>telugu_movies.csv</code> .

3.2 System Components

1. Data Loading & Preprocessing Module

Loads CSV files.
Cleans text data and standardizes formatting.
Combines multiple features into a single text column for vectorization.

2. Content Similarity Module

Uses CountVectorizer to convert text to vectors.
Computes similarity matrix using cosine similarity.

3. Activity Genre Engine

Maps moods or activities (e.g., "Date Night", "Family Time") to genre weights.
Scores movies based on genre match and average vote.

4. Filters & User Input Module

Implements Streamlit sidebar controls for filtering by:
Genre, Language, Rating, Year Range, Duration, Content Mood.

5. Recommendation Engine

Combines filtered user input with similarity scores to output movie recommendations.

6. User Interface Module

Renders movie cards, posters, titles, ratings, and detailed views.

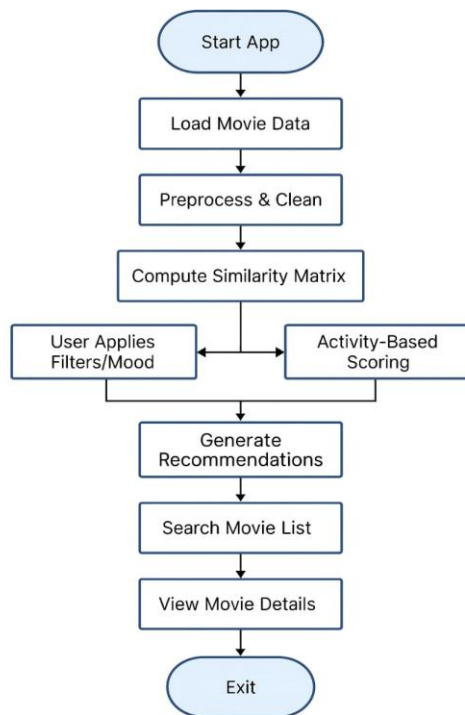
Uses modern design and interactive elements.

7. Navigation & State Management

Manages UI state across multiple views (search, details, recommendations).

Uses `st.session_state` to retain navigation context.

3.3 Flow Chart Diagram



3.4 Working Application Architecture

3.4.1 Components

Main Application Shell

Initializes session state and layout.

Loads movie data.

Decides whether to show search mode or movie detail mode.

Navigation Component

Uses the sidebar to guide interaction.

Switches views and manages filters dynamically.

3.4.2 Data Flow

Data is loaded from CSV files.

Text data is combined and vectorized.

Cosine similarity matrix is computed once and reused.

User filters trigger dynamic recommendation generation.

UI updates based on session state and user interactions.

3.4.3 API Integration

Currently uses local CSVs as data sources.

Architecture is modular and ready for future integration with APIs like TMDb or IMDb.

3.4.4 Navigation Component

Sidebar controls: Language, Rating, Year Range, Genre, Duration, Content Mood.

Session state keeps track of: Last viewed movie., Current filter selection., Navigation between views.

3.4.5 Additional Architecture Notes

Modular and scalable design.

Efficient use of session states and pre-computed similarity for performance.

Flexible code structure for future enhancements.

3.5 User Interface

3.5.1 Design Principles

Table 3.5.1: Design Principles

Principle	Description
Consistency	Uniform layout, card design, and font styling across all views.
Responsiveness	Layout adapts to screen width using Streamlit columns.
Usability	Clear filter options, sliders, and dropdowns enhance interactivity.
Aesthetics	Smooth gradient background, modern icons, and hover effects improve visual appeal.

3.5.2 Style Guide

Color Scheme

Background Gradient: #0f0f23 → #1a1a2e → #16213e

Primary Accent: #667eea → #764ba2

Highlight Colors: #ffc107 (ratings), #ff6b6b, #ff8e53 (mood-based picks)

Typography

Titles: 2.5rem, bold, white

Subtitles: 1rem, light gray

Text: 0.8rem – 1rem, consistent spacing and alignment

Icons and Graphics

Emojis: 🍿 (app icon), ⭐ (ratings), 🗣️ (language), ⌚ (duration), 📅 (year)

Star ratings: "★", "☆"

Movie cards feature hover animations and image scaling.

3.6 Additional UX Enhancements (Planned or Implemented)

🔖 Dark Mode via deep-tone gradients

🔖 Back Button in detail view to return to search results

🔖 Tooltips/Help Popups for new users (planned)

🔖 Loading Spinner for future real-time API calls

🔖 User Ratings: Thumbs-up/down for feedback (future)

4. Implementation and Results

4.1 Overview

The implementation of CinemaScope AI integrates a content-based filtering recommendation engine with a modern, responsive UI developed using Streamlit. The architecture was designed to be modular, user-friendly, and scalable for future enhancements.

- The core functionality allows users to:
- Filter movies by language, genre, rating, year, and duration
 - Get mood-based (activity-based) intelligent suggestions
 - View content-similar movies and AI-curated picks
 - Explore movie details with stylized cards and rating visualization

4.2 Front-End Components

The front end is developed using Streamlit, enhanced by custom HTML/CSS for visual appeal, interactivity, and layout management.

Table 4.2.1: Key Front-End Components

Component	Description
Main Header	Displays the app name "CinemaScope AI" with gradient styling and text shadows.
Sidebar Filters	Interactive sidebar section that includes all filter inputs such as language, duration, rating, year range, and genre.
Content Mood Selector	Dropdown to choose mood-based suggestions like “Family Night” or “Solo Evening”.
Search Bar	Allows text-based search for movie titles, genres, or keywords.
Movie Cards	Compact cards displaying movie title, genre, rating stars, and year. Includes hover animation and "Details" button.
Detail View	Expanded card for a selected movie showing all metadata (genre, rating, country, language, votes, synopsis).
Tabs Section	Displays three tabs: Similar Movies, Activity-based Recommendations, and AI Pick.
AI Pick Display	Highlights one best-rated movie from combined similarity and activity matches. Styled with red-orange gradient and boxed content.

4.3 Back-End Components

Table 4.3.1: Key Back-End Components

Component	Functionality
load_data()	Loads and combines IMDb and optional Telugu CSV data sources.
clean_data()	Cleans and processes data columns (e.g., converting vote counts, removing missing values, generating combined features).
compute_similarity_matrix()	Vectorizes combined text features (genre, description, language, country) using CountVectorizer and computes cosine_similarity.

get_content_recommendations()	Finds top N movies similar to a selected one using similarity matrix.
get_activity_recommendations()	Returns movies based on weighted genre mapping for selected activity/mood.
apply_filters()	Dynamically filters movies by user inputs (genre, rating, year, duration, language).
render_search_mode()	Displays search results or general recommendations in a grid.
render_details_mode()	Shows a full detailed view of selected movies, including all metadata and multiple recommendation sections.

4.4 Session State Management

Streamlit’s st.session_state is used extensively to manage:
User’s selected movie
Current view mode (search/details)
Filter preferences (rating, duration, language, etc.)
Search text and mood

This ensures seamless navigation and dynamic content updates without page reloads.

4.5 Key Features Implemented

Table 4.5.1: Key Features Implemented

Feature	Implementation Highlights
Dynamic Filtering	Users can filter by year, rating, genre, language, and duration using sliders and dropdowns.
Search Engine	Searches movie titles using partial match (str.contains) for real-time suggestions.
Smart Mood-Based Recommender	Matches selected moods with weighted genres and computes a custom score.
Content-Based Filtering	Computes similarity using CountVectorizer and cosine_similarity on combined textual features.
Responsive Grid Layout	Movies are displayed in a responsive 4-column layout using st.columns().
Themed UI	Gradient backgrounds, shadows, and modern CSS for a cinematic user experience.
Back Navigation	Back button is shown dynamically to allow return to search from detail view.
AI Curated Pick	Merges both content similarity and mood relevance to present the most highly rated "Smart Pick" movie.

4.6 Sample Results

Here is a sample sequence of user interaction and the resulting functionality:

1. User selects "Family Night" with filters:

Language: Telugu
Year: 2015–2020
Rating: 6.0+

Genre: All
Duration: ≤ 150 minutes

2. The app dynamically updates:

A grid of filtered movies is shown.
Clicking on a movie opens a detailed view.

Tabs display:
Movies similar to the selected one.
Movies matching “Family Night” mood.
Top pick from AI Curated logic.

3. AI Pick Example:

> AI Curated: RRR (2022)
📊 Rating: 8.7/10
Genre: Action, Adventure, Drama
“Set in the 1920s, a fictional tale about two legendary revolutionaries...”

4.7 Result Analysis

Table 4.7.1: Result Analysis

Metric	Observation
UI Responsiveness	Instant update of results and search, no page reloads required.
Accuracy	Content-based filtering shows semantically relevant movies.
Relevance	Activity-based genre mapping improves emotional and contextual relevance.
User Experience	Thematic and minimalistic UI encourages exploration and engagement.
Scalability	Can support large CSV datasets; modular structure allows easy API upgrades.

5. Technical Reference Guide

5.1 Sample Dataset Schema

A. imdb_movies.csv Columns:

Table 5.1.1: Sample Dataset Schema for imdb_movies.csv

Column Name	Description
imdb_title_id	Unique identifier for each movie (IMDB-based)
title	Title of the movie
year	Year of release
genre	Comma-separated genres
duration	Duration of the movie in minutes
country	Country of origin
language_1	Primary language (e.g., en, te)
description	Movie synopsis or overview
avg_vote	Average user rating (1–10 scale)
votes	Number of user votes received

B. telugu_movies.csv Columns (Before Mapping):

Table 5.1.2: Sample Dataset Schema for telugu_movies.csv

Original Column	Mapped To
Movie	title
Year	year
Genre	genre
Overview	description
Runtime	duration
Rating	avg_vote
No.of.Ratings	votes

5.2 Key Python Libraries Used

Table 5.2.1: Key Python Libraries Used

Library	Purpose
streamlit	Front-end rendering, UI elements, interactive layout

pandas	Data preprocessing and manipulation
numpy	Numeric calculations, array operations
sklearn.feature_extraction.text.CountVectorizer	Convert textual data into token count vectors
sklearn.metrics.pairwise.cosine_similarity	Measure similarity between movie vectors

5.3 Core Functions Overview

Table 5.3.1: Core Functions Overview

Function	Role
init_session_state()	Initializes all session-level user preferences and UI states
load_data()	Loads and combines IMDB and optional Telugu datasets
clean_data(df)	Cleans, formats, and enriches raw dataset
compute_similarity_matrix(df)	Calculates cosine similarity between movie feature vectors
get_content_recommendations()	Suggests similar movies based on content-based filtering
get_activity_recommendations()	Suggests movies based on user’s selected activity/mood
render_header()	Displays the header bar with app title and tagline
render_sidebar(df)	Displays filter options like language, year, genre, mood, etc.
render_movie_card()	Visually presents movie info in a card format
render_search_mode()	Handles the search and filter UI logic
render_details_mode()	Displays selected movie details and all three types of recommendations
main()	Entry point that calls all components in order

5.4 Activity-to-Genre Mapping

This is how user-selected activities (moods) influence genre weighting in recommendations:

Table 5.4.1: Activity-to-Genre Mapping

Activity	Genres and Weights
Solo Evening	Drama (0.4), Mystery (0.3), Thriller (0.3)
Family Night	Family (0.5), Animation (0.3), Adventure (0.2)
Date Night	Romance (0.5), Comedy (0.3), Drama (0.2)
Friends Gathering	Comedy (0.4), Action (0.3), Horror (0.3)
Learning Mode	Documentary (0.5), Biography (0.3), History (0.2)

5.5 UI Components Explained

Table 5.5.1: UI Components

Component Name	Purpose
Movie Cards	Each card displays movie title, genre, year, rating, and action button
Filter Sidebar	Allows filtering by language, genre, year, rating, duration, mood
Recommendation Tabs	Tabs include Similar, Activity-Based, and AI-Pick suggestions
Smart Pick Card	Highlights the top movie from combined content + mood matching

5.6 Screenshot Suggestion

Home Screen: Showing filters and a few movie cards.

Movie Detail View: Showing detailed info and recommendations.

AI Pick Section: Highlighting the best suggestion curated by the AI logic.

5.7 Deployment Notes

Platform: The app can be run using `streamlit run app.py`

Requirements: Include a `requirements.txt` file with:

```
streamlit
pandas
numpy
scikit-learn
```

5.8 Sample requirements.txt

```
streamlit==1.32.2
pandas==2.2.2
numpy==1.24.3
scikit-learn==1.4.2
```

6. Conclusion and Future Scope

6.1 Conclusion

The CinemaScope AI project presents a modern, intuitive, and effective approach to intelligent movie recommendation using content-based filtering, activity-driven genre mapping, and a highly interactive front-end interface built with Streamlit. By leveraging techniques such as CountVectorizer, cosine similarity, and custom activity-to-genre weight mapping, this system is capable of delivering:

- Personalized content recommendations
- Activity-oriented suggestions
- Smart filtering using multiple user preferences
- Aesthetically rich UI/UX for enhanced engagement

The platform successfully integrates multiple data sources (IMDB and Telugu movie datasets), processes them using effective natural language preprocessing, and visualizes the outputs using visually engaging cards, tabs, and thematic components.

From a usability standpoint, users benefit from:

- Simple yet powerful search capabilities
- A fully responsive filter system

Detailed views of movies including genre, duration, country, language, rating, and synopsis

Suggestions not just based on similarity but also based on user mood or context (e.g., Family Night, Learning Mode)

By embedding AI logic to blend genre preferences with average ratings, CinemaScope AI goes beyond traditional filtering—it introduces a curated and emotionally aligned viewing experience.

This makes the system especially useful for streaming platforms, movie discovery tools, and educational recommendation engines.

6.2 Future Scope

While CinemaScope AI already demonstrates a powerful proof-of-concept, several enhancements can be made to increase its scalability, accuracy, and real-world applicability.

1. Deep Learning-Based NLP

Incorporate transformer-based models (e.g., BERT, RoBERTa) to understand movie descriptions with deeper semantic meaning, outperforming the keyword-based CountVectorizer.
Implement sentence embeddings to capture contextual relationships for more accurate content similarity.

2. Hybrid Recommendation Systems

Combine collaborative filtering with content-based filtering to address the cold start problem and user preference learning over time.
Store and learn from user interactions to dynamically adapt suggestions (e.g., liked movies, watch history).

3. Integration with Real-Time APIs

Integrate with OMDB, TMDb, or Netflix APIs for live movie updates, images, trailers, and ratings.

Provide up-to-date recommendations with real-world relevance.

4. Multi-Language Expansion

Extend language support beyond English and Telugu by including movies in Hindi, Tamil, Kannada, Malayalam, etc. Use translation models for multilingual search and description matching.

5. User Profile and Login System

Implement user authentication to maintain personal profiles. Save preferences, generate history-aware suggestions, and allow rating submissions.

6. Enhanced UI Features

Add cover images/posters for movies using image APIs.
Introduce interactive charts (e.g., genre distribution, year-wise trends).
Add dark/light theme toggle for accessibility and customization.

7. Deployment and Hosting

Host the application on platforms like Streamlit Cloud, Heroku, or AWS EC2. Enable public access with secure user management for a complete SaaS prototype.

8. Mobile App Version

Build a mobile-responsive or native app (using tools like Flutter or React Native) for on-the-go movie recommendations.

In summary, CinemaScope AI lays the groundwork for a next-generation intelligent recommendation platform. With further integration of machine learning models, real-time APIs, user personalization, and scalable deployment, it has the potential to evolve into a production-grade application serving millions of users in the digital content ecosystem.

7. Bibliography

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Appendix

app.py

```
import streamlit as st
import pandas as pd
import numpy as np
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity

st.set_page_config(page_title="CinemaScope AI", page_icon=" ", layout="wide")

st.markdown("""
<style>
.stApp { background: linear-gradient(135deg, #0f0f23 0%, #1a1a2e 50%, #16213e 100%); }
.main-header {
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  padding: 1.5rem; border-radius: 12px; text-align: center; margin-bottom: 1rem;
  box-shadow: 0 10px 40px rgba(102,126,234,0.4);
}
.main-title { color: white; font-size: 2.5rem; font-weight: 800; margin: 0; text-shadow: 2px 2px 8px rgba(0,0,0,0.5); }
.main-subtitle { color: rgba(255,255,255,0.9); font-size: 1rem; margin: 0.3rem 0 0 0; font-weight: 400; }
.movie-card {
  background: linear-gradient(135deg, rgba(255,255,255,0.1) 0%, rgba(255,255,255,0.05) 100%);
  border: 1px solid rgba(102,126,234,0.3); border-radius: 10px; padding: 1rem; margin: 0.4rem 0;
  box-shadow: 0 8px 32px rgba(0,0,0,0.3); backdrop-filter: blur(15px);
  transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1); height: 180px;
}
.movie-card:hover {
  transform: translateY(-5px) scale(1.02); box-shadow: 0 15px 50px rgba(102,126,234,0.4);
  background: linear-gradient(135deg, rgba(255,255,255,0.15) 0%, rgba(255,255,255,0.1) 100%);
}
.movie-title { color: white; font-size: 0.9rem; font-weight: 700; margin-bottom: 0.4rem; line-height: 1.2; }
.movie-meta { color: #a0a6ff; font-size: 0.7rem; margin-bottom: 0.6rem; font-weight: 500; }
.rating-display {
  display: flex; align-items: center; gap: 0.4rem; margin: 0.4rem 0;
  padding: 0.2rem 0.5rem; background: rgba(102,126,234,0.2); border-radius: 5px; font-size: 0.75rem;
}
.detail-card {
  background: linear-gradient(135deg, rgba(255,255,255,0.1) 0%, rgba(255,255,255,0.05) 100%);
  border: 1px solid rgba(102,126,234,0.3); border-radius: 12px; padding: 1.5rem; margin: 1rem 0;
  box-shadow: 0 12px 40px rgba(0,0,0,0.4); backdrop-filter: blur(20px);
}
.smart-pick {
  background: linear-gradient(135deg, #ff6b6b, #ff8e53); color: white; padding: 1.2rem;
  border-radius: 12px; margin: 1rem 0; box-shadow: 0 10px 35px rgba(255,107,107,0.4);
}
.sidebar-section {
  background: rgba(255,255,255,0.1); padding: 0.8rem; border-radius: 8px; margin-bottom: 0.8rem;
  border-left: 3px solid #667eea; box-shadow: 0 4px 20px rgba(0,0,0,0.2); backdrop-filter: blur(10px);
}
.stButton > button {
  background: linear-gradient(135deg, #667eea, #764ba2); color: white; border: none;
  border-radius: 6px; padding: 0.4rem 0.8rem; font-weight: 600; font-size: 0.75rem;
  transition: all 0.3s ease; width: 100%;
}

```

```

.stButton > button:hover {
  transform: translateY(-2px); box-shadow: 0 6px 20px rgba(102,126,234,0.4);
  background: linear-gradient(135deg, #764ba2, #667eea);
}
.metric-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 1rem; margin: 1rem 0; }
.metric-item { color: white; font-size: 0.8rem; margin-bottom: 0.4rem; }
.stSelectbox > div > div, .stTextInput > div > div > input, .stSlider {
  background: rgba(255,255,255,0.1) !important; color: white !important; border: 1px solid rgba(102,126,234,0.3)
!important;
}
h1, h2, h3, h4, h5, h6, p, span, div { color: white !important; }
.stSuccess, .stWarning, .stInfo { background: rgba(255,255,255,0.1) !important; }
</style>
""", unsafe_allow_html=True)

```

```

ACTIVITY_GENRES = {
  "Solo Evening": {"Drama": 0.4, "Mystery": 0.3, "Thriller": 0.3},
  "Family Night": {"Family": 0.5, "Animation": 0.3, "Adventure": 0.2},
  "Date Night": {"Romance": 0.5, "Comedy": 0.3, "Drama": 0.2},
  "Friends Gathering": {"Comedy": 0.4, "Action": 0.3, "Horror": 0.3},
  "Learning Mode": {"Documentary": 0.5, "Biography": 0.3, "History": 0.2}
}

```

```

def init_session_state():
  defaults = {'selected_movie': None, 'view_mode': 'search', 'search_query': "",
    'current_activity': "Solo Evening", 'language': 'All', 'year_range': None,
    'rating': 6.0, 'genre': 'All', 'duration': 120}
  for key, value in defaults.items():
    if key not in st.session_state:
      st.session_state[key] = value

```

```

@st.cache_data
def load_data():
  try:
    imdb_df = pd.read_csv('imdb_movies.csv')
    columns = ['imdb_title_id', 'title', 'year', 'genre', 'duration',
      'country', 'language_1', 'description', 'avg_vote', 'votes']
    df = imdb_df[columns].copy()
    try:
      telugu_df = pd.read_csv('telugu_movies.csv')
      mapping = {'Movie': 'title', 'Year': 'year', 'Genre': 'genre', 'Overview': 'description',
        'Runtime': 'duration', 'Rating': 'avg_vote', 'No.of.Ratings': 'votes'}
      telugu_df = telugu_df.rename(columns=mapping)
      telugu_df['imdb_title_id'] = 'tl_' + telugu_df.index.astype(str)
      telugu_df['country'] = 'India'
      telugu_df['language_1'] = 'te'
      df = pd.concat([df, telugu_df[columns]], ignore_index=True)
    except FileNotFoundError:
      pass
    return clean_data(df)
  except Exception as e:
    st.error(f"Error loading data: {e}")
    return pd.DataFrame()

```

```

def clean_data(df):
  df['year'] = pd.to_numeric(df['year'], errors='coerce')
  df['duration'] = pd.to_numeric(df['duration'].astype(str).str.extract('(\d+)', expand=False), errors='coerce')

```

```

df['avg_vote'] = pd.to_numeric(df['avg_vote'], errors='coerce')
df['votes'] = pd.to_numeric(df['votes'].astype(str).str.replace(',', ''), errors='coerce')
df['description'] = df['description'].fillna('No description available')
df['genre'] = df['genre'].fillna('Unknown')
df['combined_features'] = (df['genre'] + ' ' + df['description'] + ' ' + df['country'] + ' ' + df['language_1'])
return df.drop_duplicates(subset=['title', 'year'], keep='first')

```

@st.cache_resource

def compute_similarity_matrix(df):

```

try:
    cv = CountVectorizer(max_features=3000, stop_words='english')
    vectors = cv.fit_transform(df['combined_features'].values.astype('U'))
    return cosine_similarity(vectors)
except Exception as e:
    st.error(f"Error computing similarity: {e}")
    return np.array([])

```

def get_content_recommendations(movie_title, df, similarity_matrix, num=5):

```

try:
    movie_idx = df[df['title'] == movie_title].index[0]
    scores = list(enumerate(similarity_matrix[movie_idx]))
    scores = sorted(scores, key=lambda x: x[1], reverse=True)[1:num+1]
    indices = [i[0] for i in scores]
    return df.iloc[indices]
except:
    return pd.DataFrame()

```

def get_activity_recommendations(activity, df, num=5):

```

if activity not in ACTIVITY_GENRES:
    return df.nlargest(num, 'avg_vote')
df_copy = df.copy()
df_copy['relevance_score'] = 0.0
for genre, weight in ACTIVITY_GENRES[activity].items():
    mask = df_copy['genre'].str.contains(genre, case=False, na=False)
    df_copy.loc[mask, 'relevance_score'] += weight
df_copy['combined_score'] = (0.7 * df_copy['relevance_score']) + (0.3 * df_copy['avg_vote']) / 10
return df_copy.nlargest(num, 'combined_score').drop_duplicates(subset=['title'])

```

def render_header():

```

st.markdown("""
<div class="main-header">
    <h1 class="main-title"> CinemaScope AI</h1>
    <p class="main-subtitle">Intelligent Movie Discovery Platform</p>
</div>
""", unsafe_allow_html=True)

```

def render_sidebar(df):

```

with st.sidebar:
    st.markdown('<div class="sidebar-section">', unsafe_allow_html=True)
    st.markdown('#### Filters')

    # Language Filter
    languages = {'All': 'All Languages', 'en': 'English', 'te': 'Telugu'}
    language = st.selectbox(" Language", list(languages.keys()), format_func=lambda x: languages[x])

    # Duration Filter
    duration = st.slider("⌚ Max Duration (min)", 60, 300, st.session_state.duration, 15)

```

```

st.session_state.duration = duration

# Rating Filter
rating = st.slider("📊 Min Rating", 1.0, 10.0, st.session_state.rating, 0.5)
st.session_state.rating = rating

# Year Range Filter
if st.session_state.year_range is None:
    min_year, max_year = int(df['year'].min()), int(df['year'].max())
    st.session_state.year_range = (2000, max_year)

year_range = st.slider("📅 Year Range", int(df['year'].min()), int(df['year'].max()), st.session_state.year_range)
st.session_state.year_range = year_range

# Genre Filter
all_genres = set()
for genres in df['genre'].str.split(','):
    if isinstance(genres, list):
        all_genres.update([g.strip() for g in genres])
genres = ['All'] + sorted(list(all_genres))
genre = st.selectbox("🎬 Genre", genres, index=genres.index(st.session_state.genre))
st.session_state.genre = genre
st.markdown('</div>', unsafe_allow_html=True)

# Content Mood
st.markdown('<div class="sidebar-section">', unsafe_allow_html=True)
st.markdown('#### 🎭 Content Mood')
activity = st.selectbox("Select Mood", list(ACTIVITY_GENRES.keys()),
                        index=list(ACTIVITY_GENRES.keys()).index(st.session_state.current_activity))
st.session_state.current_activity = activity
st.markdown('</div>', unsafe_allow_html=True)

if st.session_state.view_mode == 'details':
    st.markdown("---")
    if st.button("⏪ Back", use_container_width=True):
        st.session_state.view_mode = 'search'
        st.session_state.selected_movie = None
        st.rerun()

return activity, language, year_range, rating, genre, duration

def apply_filters(df, language, year_range, rating, genre, duration):
    filtered = df.copy()
    if language != 'All':
        filtered = filtered[filtered['language_1'] == language]
    filtered = filtered[
        (filtered['year'] >= year_range[0]) &
        (filtered['year'] <= year_range[1]) &
        (filtered['avg_vote'] >= rating) &
        (filtered['duration'] <= duration)
    ]
    if genre != 'All':
        filtered = filtered[filtered['genre'].str.contains(genre, na=False)]
    return filtered

def render_movie_card(movie, key_prefix):
    rating_stars = "★" * int(movie['avg_vote']/2) + "☆" * (5 - int(movie['avg_vote']/2))

```

```

st.markdown(f"""
<div class="movie-card">
    <div>
        <div class="movie-title">{movie['title']}</div>
        <div class="movie-meta">({int(movie['year'])}) • {movie['genre'][:20]}{'...' if len(movie['genre']) > 20 else
    "></div>
        <div class="rating-display">
            <span style="color: #ffc107;">{rating_stars}</span>
            <span style="font-weight: 600;">{movie['avg_vote']}/10</span>
        </div>
    </div>
</div>
""", unsafe_allow_html=True)
if st.button(" Details", key=f"{key_prefix}_{movie.name}", use_container_width=True):
    st.session_state.selected_movie = movie['title']
    st.session_state.view_mode = 'details'
    st.rerun()

def render_search_mode(df, similarity_matrix, filters):
    activity, language, year_range, rating, genre, duration = filters
    st.markdown("### Discover Movies")
    search_term = st.text_input("", placeholder="Search movies, genres, keywords...",
                                label_visibility="collapsed", value=st.session_state.search_query)
    st.session_state.search_query = search_term

    filtered_df = apply_filters(df, language, year_range, rating, genre, duration)

    if search_term and len(search_term.strip()) > 0:
        display_df = df[df['title'].str.contains(search_term, case=False, na=False)].head(20)
        if not display_df.empty:
            st.success(f"Found {len(display_df)} movies matching '{search_term}'")
        else:
            st.warning("No matches found. Showing recommendations instead.")
            display_df = filtered_df.head(20)
    else:
        display_df = filtered_df.head(20)

    if not display_df.empty:
        st.markdown("### Movies")
        cols = st.columns(4)
        for idx, (_, movie) in enumerate(display_df.iterrows()):
            with cols[idx % 4]:
                render_movie_card(movie, f"search_{idx}")
    else:
        st.info("No movies match your criteria. Try adjusting filters.")

def render_details_mode(df, similarity_matrix, activity):
    if not st.session_state.selected_movie:
        st.error("No movie selected!")
        return

    movie_data = df[df['title'] == st.session_state.selected_movie].iloc[0]

    if st.button(" Back to Discovery"):
        st.session_state.view_mode = 'search'
        st.session_state.selected_movie = None
        st.rerun()

```

```

duration_text = f"{int(movie_data['duration'])} min" if pd.notna(movie_data['duration']) else 'Unknown'
language_text = {'en': 'English', 'te': 'Telugu'}.get(movie_data['language_1'], movie_data['language_1'])
votes_text = f"{int(movie_data['votes']):,}" if pd.notna(movie_data['votes']) else 'Unknown'

st.markdown(f"""
<div class="detail-card">
  <h1 style="font-size: 1.8rem; margin-bottom: 1rem; font-weight: 700;">
    {movie_data['title']} ({int(movie_data['year'])})
  </h1>
  <div class="metric-grid">
    <div>
      <p class="metric-item"><strong> Genre:</strong> {movie_data['genre']}</p>
      <p class="metric-item"><strong>⌚ Duration:</strong> {duration_text}</p>
      <p class="metric-item"><strong> Country:</strong> {movie_data['country']}</p>
      <p class="metric-item"><strong> Language:</strong> {language_text}</p>
    </div>
    <div>
      <p class="metric-item"><strong>⭐ Rating:</strong> {movie_data['avg_vote']/10}</p>
      <p class="metric-item"><strong> Votes:</strong> {votes_text}</p>
      <div class="rating-display">
        <span style="color: #ffc107;">{'★' * int(movie_data['avg_vote']/2)}{'☆' * (5 -
int(movie_data['avg_vote']/2))}</span>
      </div>
    </div>
  </div>
  <div style="margin-top: 1rem;">
    <p style="font-weight: 600; margin-bottom: 0.5rem;"> Synopsis:</p>
    <p style="line-height: 1.4; font-size: 0.85rem;">{movie_data['description']}</p>
  </div>
</div>
""", unsafe_allow_html=True)

st.markdown("## Recommendations")
tab1, tab2, tab3 = st.tabs([" Similar", f" {activity}", " AI Pick"])

with tab1:
    similar_movies = get_content_recommendations(st.session_state.selected_movie, df, similarity_matrix, 4)
    if not similar_movies.empty:
        cols = st.columns(4)
        for idx, (_, movie) in enumerate(similar_movies.iterrows()):
            with cols[idx]:
                render_movie_card(movie, f"similar_{idx}")
    else:
        st.info("No similar movies found.")

with tab2:
    activity_movies = get_activity_recommendations(activity, df, 4)
    if not activity_movies.empty:
        cols = st.columns(4)
        for idx, (_, movie) in enumerate(activity_movies.iterrows()):
            with cols[idx]:
                render_movie_card(movie, f"activity_{idx}")

with tab3:
    similar_recs = get_content_recommendations(st.session_state.selected_movie, df, similarity_matrix, 10)
    activity_recs = get_activity_recommendations(activity, df, 10)

```

```

if not similar_recs.empty and not activity_recs.empty:
    all_movies = pd.concat([similar_recs, activity_recs]).drop_duplicates(subset=['title'])
    smart_pick = all_movies.nlargest(1, 'avg_vote').iloc[0]

    st.markdown(f"""
<div class="smart-pick">
    <h2 style="font-size: 1.3rem; margin-bottom: 0.6rem; font-weight: 600;">
        AI Curated: {smart_pick['title']}
    </h2>
    <div style="display: flex; gap: 1rem; margin-bottom: 0.6rem; font-size: 0.8rem;">
        <p><strong>👍 Rating:</strong> {smart_pick['avg_vote']}/10</p>
        <p><strong> Genre:</strong> {smart_pick['genre'][:20]}{'...' if len(smart_pick['genre']) > 20 else ''}</p>
    </div>
    <p style="font-size: 0.85rem; line-height: 1.4;">
        {smart_pick['description'][:180]}{'...' if len(smart_pick['description']) > 180 else ''}
    </p>
</div>
""", unsafe_allow_html=True)

    if st.button(" Explore AI Pick", use_container_width=True):
        st.session_state.selected_movie = smart_pick['title']
        st.rerun()
else:
    st.info("AI is analyzing preferences...")

def main():
    init_session_state()
    render_header()

    df = load_data()
    if df.empty:
        st.error("Unable to load movie database.")
        st.info("Expected files: imdb_movies.csv and optionally telugu_movies.csv")
        return

    similarity_matrix = compute_similarity_matrix(df)
    if similarity_matrix.size == 0:
        st.error("Failed to initialize recommendation engine.")
        return

    filters = render_sidebar(df)

    if st.session_state.view_mode == 'search':
        render_search_mode(df, similarity_matrix, filters)
    else:
        render_details_mode(df, similarity_matrix, filters[0])

# Run the app
if __name__ == "__main__":
    main()

```